Rosanne Liu. 2020. Plug and play language models: A simple approach to controlled text generation. In *International Conference on Learning Representations*.

Daryna Dementieva, Nikolay Babakov, and Alexander Panchenko. 2023. Detecting text formality: A study of text classification approaches. In *Proceedings of the 14th International Conference on Recent Advances in Natural Language Processing*, pages 274–284, Varna, Bulgaria. INCOMA Ltd., Shoumen, Bulgaria.

Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. 2021. A mathematical framework for transformer circuits. *Transformer Circuits Thread*. Https://transformer-circuits.pub/2021/framework/index.html.

Javier Ferrando, Gabriele Sarti, Arianna Bisazza, and Marta R. Costa-jussà. 2024. A primer on the inner workings of transformer-based language models.

Roee Hendel, Mor Geva, and Amir Globerson. 2023. In-context learning creates task vectors. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 9318–9333, Singapore. Association for Computational Linguistics.

Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. 2020. The curious case of neural text degeneration. In *International Conference on Learning Representations*.

Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. 2023. Editing models with task arithmetic. In *The Eleventh International Conference on Learning Representations*.

Jiaming Ji, Mickel Liu, Juntao Dai, Xuehai Pan, Chi Zhang, Ce Bian, Boyuan Chen, Ruiyang Sun, Yizhou Wang, and Yaodong Yang. 2023. Beavertails: Towards improved safety alignment of LLM via a human-preference dataset. In *Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.

Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, Lélio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. 2023. Mistral 7b.

Nitish Shirish Keskar, Bryan McCann, Lav R. Varshney, Caiming Xiong, and Richard Socher. 2019. Ctrl: A conditional transformer language model for controllable generation.

Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward Grefenstette, and Roberta Raileanu. 2024. Understanding the effects of RLHF on LLM generalisation and diversity. In *The Twelfth International Conference on Learning Representations*.

Ben Krause, Akhilesh Deepak Gotmare, Bryan McCann, Nitish Shirish Keskar, Shafiq Joty, Richard Socher, and Nazneen Fatema Rajani. 2021. GeDi: Generative discriminator guided sequence generation. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, pages 4929–4952, Punta Cana, Dominican Republic. Association for Computational Linguistics.

Chak Leong, Yi Cheng, Jiashuo Wang, Jian Wang, and Wenjie Li. 2023. Self-detoxifying language models via toxification reversal. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 4433–4449, Singapore. Association for Computational Linguistics.

Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. 2023a. Inference-time intervention: Eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems*, volume 36, pages 41451–41530. Curran Associates, Inc.

Xiang Lisa Li, Ari Holtzman, Daniel Fried, Percy Liang, Jason Eisner, Tatsunori Hashimoto, Luke Zettlemoyer, and Mike Lewis. 2023b. Contrastive decoding: Open-ended text generation as optimization. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12286–12312, Toronto, Canada. Association for Computational Linguistics.

Xiang Lisa Li and Percy Liang. 2021. Prefix-tuning: Optimizing continuous prompts for generation. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 4582–4597, Online. Association for Computational Linguistics.

Alisa Liu, Maarten Sap, Ximing Lu, Swabha Swayamdipta, Chandra Bhagavatula, Noah A. Smith, and Yejin Choi. 2021. DExperts: Decoding-time controlled text generation with experts and anti-experts. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 6691–6706, Online. Association for Computational Linguistics.

Sheng Liu, Haotian Ye, Lei Xing, and James Zou. 2023. In-context vectors: Making in context learning more effective and controllable through latent space steering.

Samuel Marks and Max Tegmark. 2023. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets.

Tomas Mikolov, Wen-tau Yih, and Geoffrey Zweig. 2013. Linguistic regularities in continuous space word representations. In *Proceedings of the 2013 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 746–751, Atlanta, Georgia. Association for Computational Linguistics.

Shuyo Nakatani. 2010. Language detection library for java.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul F Christiano, Jan Leike, and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pages 27730–27744. Curran Associates, Inc.

Kiho Park, Yo Joong Choe, and Victor Veitch. 2023. The linear representation hypothesis and the geometry of large language models.

Sudha Rao and Joel Tetreault. 2018. Dear sir or madam, may I introduce the GYAFC dataset: Corpus, benchmarks and metrics for formality style transfer. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 129–140, New Orleans, Louisiana. Association for Computational Linguistics.

Emily Reif, Daphne Ippolito, Ann Yuan, Andy Coenen, Chris Callison-Burch, and Jason Wei. 2022. A recipe for arbitrary text style transfer with large language models. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 837–848, Dublin, Ireland. Association for Computational Linguistics.

Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. 2024. Steering llama 2 via contrastive activation addition.

Gabriele Sarti, Grzegorz Chrupała, Malvina Nissim, and Arianna Bisazza. 2024. Quantifying the plausibility of context reliance in neural machine translation. In *The Twelfth International Conference on Learning Representations (ICLR 2024)*, Vienna, Austria. OpenReview.

Gabriele Sarti, Phu Mon Htut, Xing Niu, Benjamin Hsu, Anna Currey, Georgiana Dinu, and Maria Nadejde. 2023. RAMP: Retrieval and attribute-marking enhanced prompting for attribute-controlled translation. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 1476–1490, Toronto, Canada. Association for Computational Linguistics.

Nishant Subramani, Nivedita Suresh, and Matthew Peters. 2022. Extracting latent steering vectors from pretrained language models. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 566–581, Dublin, Ireland. Association for Computational Linguistics.

Mirac Suzgun, Luke Melas-Kyriazi, and Dan Jurafsky. 2022. Prompt-and-rerank: A method for zero-shot and few-shot arbitrary textual style transfer with small language models. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 2195–2222, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. 2023. Stanford alpaca: An instruction-following llama model. https://github.com/tatsu-lab/stanford_alpaca.

NLLB Team, Marta R. Costa-jussà, James Cross, Onur Çelebi, Maha Elbayad, Kenneth Heafield, Kevin Heffernan, Elahe Kalbassi, Janice Lam, Daniel Licht, Jean Maillard, Anna Sun, Skyler Wang, Guillaume Wenzek, Al Youngblood, Bapi Akula, Loic Barrault, Gabriel Mejia Gonzalez, Prangthip Hansanti, John Hoffman, Semarley Jarrett, Kaushik Ram Sadagopan, Dirk Rowe, Shannon Spruit, Chau Tran, Pierre Andrews, Necip Fazil Ayan, Shruti Bhosale, Sergey Edunov, Angela Fan, Cynthia Gao, Vedanuj Goswami, Francisco Guzmán, Philipp Koehn, Alexandre Mourachko, Christophe Ropers, Safiyyah Saleem, Holger Schwenk, and Jeff Wang. 2022. No language left behind: Scaling human-centered machine translation.

Curt Tigges, Oskar John Hollinsworth, Atticus Geiger, and Neel Nanda. 2023. Linear representations of sentiment in large language models.

Eric Todd, Millicent Li, Arnab Sen Sharma, Aaron Mueller, Byron C Wallace, and David Bau. 2024. Function vectors in large language models. In *The Twelfth International Conference on Learning Representations*.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. 2023. Activation addition: Steering language models without optimization.

Jannis Vamvas and Rico Sennrich. 2021. Contrastive conditioning for assessing disambiguation in MT: A case study of distilled bias. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 10246–10265, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.

Jannis Vamvas and Rico Sennrich. 2022. As little as possible, as much as necessary: Detecting over- and undertranslations with contrastive conditioning. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 490–500, Dublin, Ireland. Association for Computational Linguistics.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Ł ukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc.

Kevin Yang and Dan Klein. 2021. FUDGE: Controlled text generation with future discriminators. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 3511–3535, Online. Association for Computational Linguistics.

Daniel M. Ziegler, Nisan Stiennon, Jeffrey Wu, Tom B. Brown, Alec Radford, Dario Amodei, Paul Christiano, and Geoffrey Irving. 2019. Fine-tuning language models from human preferences.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. 2023. Representation engineering: A top-down approach to ai transparency.

A Additional Background

A.1 Attention Activations in Transformer Language Models

The generic structure of a language model with transformers architecture (Vaswani et al., 2017) starts with an embedding procedure where each token of the prompt $p = \langle t_1, \ldots, t_n \rangle$ is transformed in a sequence of embeddings $x = \langle x_1, \ldots, x_n \rangle$ where $x \in \mathbb{R}^d$ with d being the embedding dimension. The prompt representation is fed to the model as $f(x)$ which is trained to return the next predicted token x_{n+1}. By following the Elhage et al. (2021) perspective on the transformer architecture, we define $X^l \in \mathbb{R}^{n \times d}$ as the layer $l \in L$ internal representation of the model's input.

Each layer includes different components that operate in sequence on the internal representation X^l keeping a residual connection from the previous state:

$$X^l = X^{mid} + \text{MLP}^l(X^{mid}) \tag{7}$$

with MLP being a fully connected feed-forward network at the l-th layer and X^{mid} defined as:

$$X^{\text{mid}} = X^{l-1} + \sum^{H} Attn^{l,h}(X^{l-1}) \tag{8}$$

One fundamental component in auto-regressive transformer models is the attention block $Attn$ which helps the model contextualize each token representation X_i^{l-1} to its previous token representations $X_{\leq i}^{l-1}$, eventually writing the final output to the current residual stream X^l.

To this end, the residual stream X^{l-1} is split across the total number of attention heads H in the transformer architecture. Each h-th attention head computes its output as follows:

$$Attn^{l,h}(X_{\leq i}^{l-1}) = \sum_{j=0}^{i} a_{i,j}^{l,h} x_j^{l-1} W_V^{l,h} W_O^{l,h} \tag{9}$$

with $W_V^{l,h}$ and $W_O^{l,h}$ being the output and value learnable parameters and $a_i^{l,h}$ defined as:

$$a_i^{l,h} = \text{softmax}\left(\frac{x_i^{l-1} W_Q^{l,h} (X_{\leq i}^{l-1} W_K^{l,h})^\intercal}{\sqrt{d_k}} \right) \tag{10}$$

where $W_Q^{l,h}$ and $W_K^{l,h}$ are the query and key parameters. Our framework focuses on the last token representation of the prompt x_n from the attention output. For this reason, we define $v^{l,h}$ as the output activation from the attention mechanism for each head h for each layer l as follows:

$$v^{l,h} = Attn^{l,h}(X_n^{l-1}) \tag{11}$$

The last residual stream x^L is converted to a next-token distribution of logits $\mathcal{V}$ through the unembedding matrix W_u which will be used to get the next predicted token following the initial prompt.

$$f(x) = x^L W_u = \mathcal{V}$$

where $\mathcal{V} \in \mathbb{R}^{d \times ||\mathcal{V}||}$ with $||\mathcal{V}||$ being the vocabulary dimension of the model. Finally the predicted token y_0 is obtained with $y_0 = \text{argmax}(\mathcal{V})$.

A.2 Activation Steering Approaches

Several aspects in common and not in common with previous works on the same subject are briefly addressed below.